Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

April 28, 2023

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50342119

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on April 14, 2023. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

CERTIFICATIONS

Project:	Biosolids
Pace Project No.:	50342119

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019
Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Foreign Soil Permit #: 525-23-13-23119
USDA Compliance Agreement #: IN-SL-22-001

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SAMPLE SUMMARY

Project: Biosolids
Pace Project No.: 50342119

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50342119001	**Digestor #3**	Solid	04/13/23 14:20	04/14/23 10:40

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50342119

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50342119001	**Digestor #3**	EPA 8081	KAV	21	PASI-I
		EPA 8081	KAV	8	PASI-I
		EPA 8082	AM	8	PASI-I
		EPA 8151A	CPH	3	PASI-I
		EPA 6010	DJS	6	PASI-I
		EPA 6010	JPK	8	PASI-I
		EPA 6020	MGM	10	PASI-I
		EPA 7470	EAE	1	PASI-I
		EPA 7471	ILP	1	PASI-I
		EPA 8270	JCM	67	PASI-I
		EPA 8270	JCM	18	PASI-I
		EPA 8260	KLP, TMW	55	PASI-I
		EPA 5030/8260	BES	13	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	QAK	1	PASI-I
		EPA 9038	BEP	1	PASI-I
		EPA 9045	BMS	1	PASI-I
		EPA 351.2	OAS	1	PASI-I
		EPA 353.2	ZM	2	PASI-I
		SM 4500-NH3 G	MMS	1	PASI-I
		SM 4500-Cl-E	ZM	1	PASI-I
		SM 4500-CN-E	MMS	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids	Analytical Method: EPA 8081 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Aldrin	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	309-00-2	
alpha-BHC	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	319-84-6	
beta-BHC	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	319-85-7	
delta-BHC	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	319-86-8	
gamma-BHC (Lindane)	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	58-89-9	
Chlordane (Technical)	**<5880**	ug/kg	5880	1	04/17/23 15:17	04/25/23 20:52	57-74-9	
alpha-Chlordane	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	5103-71-9	
gamma-Chlordane	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	5103-74-2	
4,4'-DDD	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	72-54-8	
4,4'-DDE	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	72-55-9	
4,4'-DDT	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	50-29-3	
Dieldrin	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	60-57-1	
Endosulfan I	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	959-98-8	
Endosulfan II	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	33213-65-9	
Endosulfan sulfate	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	1031-07-8	
Endrin	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	72-20-8	
Endrin aldehyde	**<588**	ug/kg	588	1	04/17/23 15:17	04/25/23 20:52	7421-93-4	
Heptachlor	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	76-44-8	
Heptachlor epoxide	**<294**	ug/kg	294	1	04/17/23 15:17	04/25/23 20:52	1024-57-3	
Toxaphene	**<5880**	ug/kg	5880	1	04/17/23 15:17	04/25/23 20:52	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	39	%.	10-166	1	04/17/23 15:17	04/25/23 20:52	2051-24-3	
8081 GCS Pest RV, TCLP	Analytical Method: EPA 8081 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 04/21/23 15:30 Initial pH: 8.68; Final pH: 5.2							
	Pace Analytical Services - Indianapolis							
gamma-BHC (Lindane)	**<0.00025**	mg/L	0.00025	1	04/25/23 08:30	04/25/23 18:58	58-89-9	
Chlordane (Technical)	**<0.0050**	mg/L	0.0050	1	04/25/23 08:30	04/25/23 18:58	57-74-9	
Endrin	**<0.00050**	mg/L	0.00050	1	04/25/23 08:30	04/25/23 18:58	72-20-8	
Heptachlor	**<0.00025**	mg/L	0.00025	1	04/25/23 08:30	04/25/23 18:58	76-44-8	
Heptachlor epoxide	**<0.00025**	mg/L	0.00025	1	04/25/23 08:30	04/25/23 18:58	1024-57-3	
Methoxychlor	**<0.0025**	mg/L	0.0025	1	04/25/23 08:30	04/25/23 18:58	72-43-5	
Toxaphene	**<0.0050**	mg/L	0.0050	1	04/25/23 08:30	04/25/23 18:58	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	29	%.	10-124	1	04/25/23 08:30	04/25/23 18:58	2051-24-3	
8082 PCB Solids	Analytical Method: EPA 8082 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
PCB-1016 (Aroclor 1016)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	12674-11-2	
PCB-1221 (Aroclor 1221)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	11104-28-2	
PCB-1232 (Aroclor 1232)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	11141-16-5	
PCB-1242 (Aroclor 1242)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	53469-21-9	
PCB-1248 (Aroclor 1248)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	12672-29-6	
PCB-1254 (Aroclor 1254)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	11097-69-1	
PCB-1260 (Aroclor 1260)	**<9470**	ug/kg	9470	1	04/19/23 10:36	04/24/23 21:13	11096-82-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8082 PCB Solids	Analytical Method: EPA 8082 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Surrogates								
Tetrachloro-m-xylene (S)	28	%.	10-133	1	04/19/23 10:36	04/24/23 21:13	877-09-8	
8151A Cl Acid Herbicides TCLP	Analytical Method: EPA 8151A Preparation Method: EPA 8151							
	Leachate Method/Date: EPA 1311; 04/21/23 15:30 Initial pH: 8.68; Final pH: 5.2							
	Pace Analytical Services - Indianapolis							
2,4-D	**<0.0050**	mg/L	0.0050	1	04/25/23 11:16	04/27/23 19:53	94-75-7	
2,4,5-TP (Silvex)	**<0.0050**	mg/L	0.0050	1	04/25/23 11:16	04/27/23 19:53	93-72-1	
Surrogates								
2,4-DCAA (S)	68	%.	22-132	1	04/25/23 11:16	04/27/23 20:11	19719-28-9	
6010 MET ICP	Analytical Method: EPA 6010 Preparation Method: EPA 3050							
	Pace Analytical Services - Indianapolis							
Aluminum	**8490**	mg/kg	1600	1	04/25/23 15:08	04/26/23 13:36	7429-90-5	
Calcium	**25100**	mg/kg	1600	1	04/25/23 15:08	04/26/23 13:36	7440-70-2	
Magnesium	**5500**	mg/kg	1600	1	04/25/23 15:08	04/26/23 13:36	7439-95-4	
Phosphorus	**28500**	mg/kg	1600	1	04/25/23 15:08	04/26/23 13:36	7723-14-0	N2
Potassium	**2860**	mg/kg	1600	1	04/25/23 15:08	04/26/23 13:36	7440-09-7	
Sodium	**3970**	mg/kg	1600	1	04/25/23 15:08	04/26/23 13:36	7440-23-5	
6010 MET ICP, TCLP	Analytical Method: EPA 6010 Preparation Method: EPA 3010							
	Leachate Method/Date: EPA 1311; 04/21/23 15:30 Initial pH: 8.68; Final pH: 5.2							
	Pace Analytical Services - Indianapolis							
Arsenic	**<0.10**	mg/L	0.10	1	04/24/23 08:10	04/24/23 23:19	7440-38-2	
Barium	**<5.0**	mg/L	5.0	1	04/24/23 08:10	04/24/23 23:19	7440-39-3	
Cadmium	**<0.050**	mg/L	0.050	1	04/24/23 08:10	04/24/23 23:19	7440-43-9	
Chromium	**<0.10**	mg/L	0.10	1	04/24/23 08:10	04/24/23 23:19	7440-47-3	
Lead	**<0.10**	mg/L	0.10	1	04/24/23 08:10	04/24/23 23:19	7439-92-1	
Selenium	**<0.10**	mg/L	0.10	1	04/24/23 08:10	04/24/23 23:19	7782-49-2	
Silver	**<0.10**	mg/L	0.10	1	04/24/23 08:10	04/24/23 23:19	7440-22-4	
Zinc	**3.0**	mg/L	0.20	1	04/24/23 08:10	04/24/23 23:19	7440-66-6	
6020 MET ICPMS	Analytical Method: EPA 6020 Preparation Method: EPA 3050B							
	Pace Analytical Services - Indianapolis							
Arsenic	**12.4**	mg/kg	0.56	1	04/25/23 21:19	04/26/23 20:15	7440-38-2	
Cadmium	**1.6**	mg/kg	0.28	1	04/25/23 21:19	04/26/23 20:15	7440-43-9	
Chromium	**75.9**	mg/kg	1.1	1	04/25/23 21:19	04/26/23 20:15	7440-47-3	
Copper	**505**	mg/kg	2.8	5	04/25/23 21:19	04/26/23 20:11	7440-50-8	
Lead	**15.2**	mg/kg	2.8	5	04/25/23 21:19	04/26/23 20:11	7439-92-1	
Molybdenum	**11.7**	mg/kg	0.56	1	04/25/23 21:19	04/26/23 20:15	7439-98-7	N2
Nickel	**15.2**	mg/kg	0.28	1	04/25/23 21:19	04/26/23 20:15	7440-02-0	
Selenium	**5.9**	mg/kg	0.56	1	04/25/23 21:19	04/26/23 20:15	7782-49-2	
Silver	**1.5**	mg/kg	0.28	1	04/25/23 21:19	04/26/23 20:15	7440-22-4	
Zinc	**899**	mg/kg	28.2	10	04/25/23 21:19	04/26/23 20:07	7440-66-6	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP	Analytical Method: EPA 7470 Preparation Method: EPA 7470 Leachate Method/Date: EPA 1311; 04/21/23 15:30 Initial pH: 8.68; Final pH: 5.2 Pace Analytical Services - Indianapolis							
Mercury	**<0.0020**	mg/L	0.0020	1	04/25/23 10:02	04/25/23 17:07	7439-97-6	
7471 Mercury	Analytical Method: EPA 7471 Preparation Method: EPA 7471 Pace Analytical Services - Indianapolis							
Mercury	**<7.2**	mg/kg	7.2	1	04/19/23 19:30	04/20/23 10:24	7439-97-6	
8270 SVOC SS Soil	Analytical Method: EPA 8270 Preparation Method: EPA 3546 Pace Analytical Services - Indianapolis							
Acenaphthene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	83-32-9	
Acenaphthylene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	208-96-8	
Anthracene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	120-12-7	
Benzo(a)anthracene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	56-55-3	
Benzo(a)pyrene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	50-32-8	
Benzo(b)fluoranthene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	205-99-2	
Benzo(g,h,i)perylene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	191-24-2	
Benzo(k)fluoranthene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	207-08-9	
Benzyl alcohol	**<84400**	ug/kg	84400	1	04/19/23 12:17	04/19/23 22:54	100-51-6	
4-Bromophenylphenyl ether	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	101-55-3	
Butylbenzylphthalate	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	85-68-7	
4-Chloro-3-methylphenol	**<84400**	ug/kg	84400	1	04/19/23 12:17	04/19/23 22:54	59-50-7	
4-Chloroaniline	**<84400**	ug/kg	84400	1	04/19/23 12:17	04/19/23 22:54	106-47-8	
bis(2-Chloroethoxy)methane	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	111-91-1	
bis(2-Chloroethyl) ether	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	111-44-4	
bis(2chloro1methylethyl) ether	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	108-60-1	
2-Chloronaphthalene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	91-58-7	
2-Chlorophenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	95-57-8	
4-Chlorophenylphenyl ether	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	7005-72-3	
Chrysene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	218-01-9	
Dibenz(a,h)anthracene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	53-70-3	
Dibenzofuran	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	132-64-9	
3,3'-Dichlorobenzidine	**<84400**	ug/kg	84400	1	04/19/23 12:17	04/19/23 22:54	91-94-1	
2,4-Dichlorophenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	120-83-2	
Diethylphthalate	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	84-66-2	
2,4-Dimethylphenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	105-67-9	
Dimethylphthalate	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	131-11-3	
Di-n-butylphthalate	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	84-74-2	
4,6-Dinitro-2-methylphenol	**<84400**	ug/kg	84400	1	04/19/23 12:17	04/19/23 22:54	534-52-1	
2,4-Dinitrophenol	**<204000**	ug/kg	204000	1	04/19/23 12:17	04/19/23 22:54	51-28-5	
2,4-Dinitrotoluene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	121-14-2	
2,6-Dinitrotoluene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	606-20-2	
Di-n-octylphthalate	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	117-84-0	
bis(2-Ethylhexyl)phthalate	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	117-81-7	
Fluoranthene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	206-44-0	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil	Analytical Method: EPA 8270 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Fluorene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	86-73-7	
Hexachloro-1,3-butadiene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	87-68-3	
Hexachlorobenzene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	118-74-1	
Hexachlorocyclopentadiene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	77-47-4	
Hexachloroethane	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	67-72-1	
Indeno(1,2,3-cd)pyrene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	193-39-5	
Isophorone	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	78-59-1	
1-Methylnaphthalene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	90-12-0	
2-Methylnaphthalene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	91-57-6	
2-Methylphenol(o-Cresol)	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	95-48-7	
3&4-Methylphenol(m&p Cresol)	**<84400**	ug/kg	84400	1	04/19/23 12:17	04/19/23 22:54		
Naphthalene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	91-20-3	
2-Nitroaniline	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	88-74-4	
3-Nitroaniline	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	99-09-2	
4-Nitroaniline	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	100-01-6	
Nitrobenzene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	98-95-3	
2-Nitrophenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	88-75-5	
4-Nitrophenol	**<204000**	ug/kg	204000	1	04/19/23 12:17	04/19/23 22:54	100-02-7	
N-Nitroso-di-n-propylamine	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	621-64-7	
N-Nitrosodiphenylamine	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	86-30-6	
Pentachlorophenol	**<204000**	ug/kg	204000	1	04/19/23 12:17	04/19/23 22:54	87-86-5	
Phenanthrene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	85-01-8	
Phenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	108-95-2	
Pyrene	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	129-00-0	
2,4,5-Trichlorophenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	95-95-4	
2,4,6-Trichlorophenol	**<42200**	ug/kg	42200	1	04/19/23 12:17	04/19/23 22:54	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	55	%.	35-110	1	04/19/23 12:17	04/19/23 22:54	4165-60-0	
Phenol-d5 (S)	64	%.	35-115	1	04/19/23 12:17	04/19/23 22:54	4165-62-2	
2-Fluorophenol (S)	66	%.	22-114	1	04/19/23 12:17	04/19/23 22:54	367-12-4	
2,4,6-Tribromophenol (S)	76	%.	10-123	1	04/19/23 12:17	04/19/23 22:54	118-79-6	
2-Fluorobiphenyl (S)	69	%.	36-100	1	04/19/23 12:17	04/19/23 22:54	321-60-8	
p-Terphenyl-d14 (S)	78	%.	29-117	1	04/19/23 12:17	04/19/23 22:54	1718-51-0	
8270 MSSV TCLP Sep Funnel	Analytical Method: EPA 8270 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 04/21/23 15:30 Initial pH: 8.68; Final pH: 5.2							
	Pace Analytical Services - Indianapolis							
1,4-Dichlorobenzene	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	106-46-7	
2,4-Dinitrotoluene	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	121-14-2	
Hexachloro-1,3-butadiene	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	87-68-3	
Hexachlorobenzene	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	118-74-1	
Hexachloroethane	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	67-72-1	
2-Methylphenol(o-Cresol)	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	95-48-7	
3&4-Methylphenol(m&p Cresol)	**<0.20**	mg/L	0.20	1	04/26/23 13:43	04/27/23 20:36		
Nitrobenzene	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	98-95-3	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 MSSV TCLP Sep Funnel	Analytical Method: EPA 8270 Preparation Method: EPA 3510 Leachate Method/Date: EPA 1311; 04/21/23 15:30 Initial pH: 8.68; Final pH: 5.2 Pace Analytical Services - Indianapolis							
Pentachlorophenol	**<0.50**	mg/L	0.50	1	04/26/23 13:43	04/27/23 20:36	87-86-5	
Pyridine	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	110-86-1	
2,4,5-Trichlorophenol	**<0.50**	mg/L	0.50	1	04/26/23 13:43	04/27/23 20:36	95-95-4	
2,4,6-Trichlorophenol	**<0.10**	mg/L	0.10	1	04/26/23 13:43	04/27/23 20:36	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	59	%.	44-107	1	04/26/23 13:43	04/27/23 20:36	4165-60-0	
2-Fluorobiphenyl (S)	56	%.	36-92	1	04/26/23 13:43	04/27/23 20:36	321-60-8	
p-Terphenyl-d14 (S)	95	%.	57-146	1	04/26/23 13:43	04/27/23 20:36	1718-51-0	
Phenol-d5 (S)	25	%.	15-47	1	04/26/23 13:43	04/27/23 20:36	4165-62-2	
2-Fluorophenol (S)	38	%.	23-69	1	04/26/23 13:43	04/27/23 20:36	367-12-4	
2,4,6-Tribromophenol (S)	109	%.	37-127	1	04/26/23 13:43	04/27/23 20:36	118-79-6	
8260 MSV 5030 Low Level	Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis							
Acetone	**5.0**	mg/kg	3.1	1		04/24/23 17:37	67-64-1	
Acrolein	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	107-02-8	
Acrylonitrile	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	107-13-1	
Benzene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	71-43-2	
Bromodichloromethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-27-4	
Bromoform	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-25-2	
Bromomethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	74-83-9	
2-Butanone (MEK)	**<0.82**	mg/kg	0.82	1		04/21/23 02:41	78-93-3	
Carbon disulfide	**<0.33**	mg/kg	0.33	1		04/21/23 02:41	75-15-0	
Carbon tetrachloride	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	56-23-5	
Chlorobenzene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	108-90-7	
Chloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-00-3	L2
Chloroform	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	67-66-3	
Chloromethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	74-87-3	
1,2-Dibromo-3-chloropropane	**<0.33**	mg/kg	0.33	1		04/21/23 02:41	96-12-8	
Dibromochloromethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	124-48-1	
1,2-Dibromoethane (EDB)	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	106-93-4	
Dibromomethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	74-95-3	
1,2-Dichlorobenzene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	95-50-1	
1,3-Dichlorobenzene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	541-73-1	
1,4-Dichlorobenzene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	106-46-7	
trans-1,4-Dichloro-2-butene	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	110-57-6	
Dichlorodifluoromethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-71-8	
1,1-Dichloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-34-3	
1,2-Dichloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	107-06-2	
1,1-Dichloroethene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-35-4	
cis-1,2-Dichloroethene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	156-59-2	
trans-1,2-Dichloroethene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	156-60-5	
1,2-Dichloropropane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	78-87-5	
cis-1,3-Dichloropropene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	10061-01-5	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level	Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis							
trans-1,3-Dichloropropene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	10061-02-6	
Ethylbenzene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	100-41-4	
Ethyl methacrylate	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	97-63-2	
2-Hexanone	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	591-78-6	
Iodomethane	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	74-88-4	
Methylene Chloride	**<0.65**	mg/kg	0.65	1		04/21/23 02:41	75-09-2	
Methyl methacrylate	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	80-62-6	
4-Methyl-2-pentanone (MIBK)	**<0.82**	mg/kg	0.82	1		04/21/23 02:41	108-10-1	
Methyl-tert-butyl ether	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	1634-04-4	
Styrene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	100-42-5	
1,1,1,2-Tetrachloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	630-20-6	
1,1,2,2-Tetrachloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	79-34-5	
Tetrachloroethene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	127-18-4	
Toluene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	108-88-3	
1,1,1-Trichloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	71-55-6	
1,1,2-Trichloroethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	79-00-5	
Trichloroethene	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	79-01-6	
Trichlorofluoromethane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-69-4	
1,2,3-Trichloropropane	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	96-18-4	
Vinyl acetate	**<3.3**	mg/kg	3.3	1		04/21/23 02:41	108-05-4	
Vinyl chloride	**<0.16**	mg/kg	0.16	1		04/21/23 02:41	75-01-4	
Xylene (Total)	**<0.33**	mg/kg	0.33	1		04/21/23 02:41	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	101	%.	75-135	1		04/21/23 02:41	1868-53-7	
Toluene-d8 (S)	105	%.	65-148	1		04/21/23 02:41	2037-26-5	
4-Bromofluorobenzene (S)	93	%.	63-132	1		04/21/23 02:41	460-00-4	
8260 MSV TCLP	Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 04/25/23 16:00 Pace Analytical Services - Indianapolis							
Benzene	**<0.050**	mg/L	0.050	1		04/27/23 12:34	71-43-2	
2-Butanone (MEK)	**<1.0**	mg/L	1.0	1		04/27/23 12:34	78-93-3	
Carbon tetrachloride	**<0.050**	mg/L	0.050	1		04/27/23 12:34	56-23-5	
Chlorobenzene	**<0.050**	mg/L	0.050	1		04/27/23 12:34	108-90-7	
Chloroform	**<0.050**	mg/L	0.050	1		04/27/23 12:34	67-66-3	
1,2-Dichloroethane	**<0.050**	mg/L	0.050	1		04/27/23 12:34	107-06-2	
1,1-Dichloroethene	**<0.050**	mg/L	0.050	1		04/27/23 12:34	75-35-4	
Tetrachloroethene	**<0.050**	mg/L	0.050	1		04/27/23 12:34	127-18-4	
Trichloroethene	**<0.050**	mg/L	0.050	1		04/27/23 12:34	79-01-6	
Vinyl chloride	**<0.020**	mg/L	0.020	1		04/27/23 12:34	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	100	%.	79-124	1		04/27/23 12:34	460-00-4	
Dibromofluoromethane (S)	102	%.	82-128	1		04/27/23 12:34	1868-53-7	
Toluene-d8 (S)	99	%.	73-122	1		04/27/23 12:34	2037-26-5	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50342119

Sample: Digestor #3 **Lab ID: 50342119001** Collected: 04/13/23 14:20 Received: 04/14/23 10:40 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
Percent Moisture	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Percent Moisture	**97.1**	%	0.10	1		04/17/23 14:40		N2
2540G Total Percent Solids	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Total Solids	**2.9**	%	0.10	1		04/17/23 15:01		N2
2540G Total Volatile Solids	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Total Volatile Solids	**46.7**	%	0.10	1		04/17/23 14:40		N2
9038 Sulfate Soil	Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis							
Sulfate	**<3410**	mg/kg	3410	1	04/26/23 09:34	04/26/23 11:13	14808-79-8	N2,P4
9045 pH Soil	Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis							
pH at 25 Degrees C	**7.9**	Std. Units	0.10	1		04/24/23 10:21		H3
351.2 Total Kjeldahl Nitrogen	Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis							
Nitrogen, Kjeldahl, Total	**54400**	mg/kg	1630	1	04/20/23 08:11	04/21/23 13:52	7727-37-9	N2
353.2 Nitrogen, NO2/NO3	Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis							
Nitrogen, NO2 plus NO3	**<171**	mg/kg	171	1	04/26/23 11:41	04/26/23 22:52		N2
Nitrogen, Nitrate	**<171**	mg/kg	171	1	04/26/23 11:41	04/26/23 22:52	14797-55-8	N2
4500 Ammonia Soil, Distilled	Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis							
Nitrogen, Ammonia	**37900**	mg/kg	1440	1	04/17/23 11:10	04/17/23 14:28	7664-41-7	N2
4500 Chloride in Soil	Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis							
Chloride	**4240**	mg/kg	3420	1	04/26/23 11:00	04/27/23 13:07	16887-00-6	N2
4500CNE Cyanide, Soil	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis							
Cyanide	**<16.5**	mg/kg	16.5	1	04/27/23 08:32	04/27/23 14:50	57-12-5	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729481
QC Batch Method: EPA 7470
Analysis Method: EPA 7470
Analysis Description: 7470 Mercury TCLP
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50342119001

METHOD BLANK: 3347923 Matrix: Water
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	04/25/23 16:38	

LABORATORY CONTROL SAMPLE: 3347924

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0053	106	80-120	

MATRIX SPIKE SAMPLE: 3347925

Parameter	Units	50342006001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.15	0.16	107	75-125	

MATRIX SPIKE SAMPLE: 3347926

Parameter	Units	50342150001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.03	0.033	109	75-125	

MATRIX SPIKE SAMPLE: 3347927

Parameter	Units	50342437001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	107	75-125	

MATRIX SPIKE SAMPLE: 3347928

Parameter	Units	50342113001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.016	106	75-125	

MATRIX SPIKE SAMPLE: 3347929

Parameter	Units	50342119001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.014	95	75-125	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

MATRIX SPIKE SAMPLE: 3347930

Parameter	Units	50342146001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	<0.0020	10	75-125	M0

MATRIX SPIKE SAMPLE: 3347931

Parameter	Units	50342248001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	102	75-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3347932 3347933

Parameter	Units	50342569002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	ND	0.015	0.015	0.016	0.016	106	104	75-125	2	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 728535
QC Batch Method: EPA 7471
Analysis Method: EPA 7471
Analysis Description: 7471 Mercury
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50342119001

METHOD BLANK: 3343264 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	04/20/23 09:16	

LABORATORY CONTROL SAMPLE: 3343265

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.53	105	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3343266 3343267

Parameter	Units	50341810006 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.61	0.62	0.74	0.93	113	140	75-125	23	20	M0,R1

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729313
QC Batch Method: EPA 3050
Analysis Method: EPA 6010
Analysis Description: 6010 MET
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3347016 Matrix: Solid

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	04/26/23 11:47	
Calcium	mg/kg	<50.0	50.0	04/26/23 11:47	
Magnesium	mg/kg	<50.0	50.0	04/26/23 11:47	
Phosphorus	mg/kg	ND	50.0	04/26/23 11:47	N2
Potassium	mg/kg	<50.0	50.0	04/26/23 11:47	
Sodium	mg/kg	<50.0	50.0	04/26/23 11:47	

LABORATORY CONTROL SAMPLE: 3347017

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	478	96	80-120	
Calcium	mg/kg	500	510	102	80-120	
Magnesium	mg/kg	500	488	98	80-120	
Phosphorus	mg/kg	50	<50.0	98	80-120	N2
Potassium	mg/kg	500	496	99	80-120	
Sodium	mg/kg	500	492	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3347018 3347019

Parameter	Units	50342246003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	14500	894	907	21700	17800	803	369	75-125	19	20	P6
Calcium	mg/kg	20300	894	907	24200	18800	430	-171	75-125	25	20	E,P6, R1
Magnesium	mg/kg	6390	894	907	8160	6800	198	46	75-125	18	20	P6
Phosphorus	mg/kg	808	89.4	90.7	878	760	79	-53	75-125	14	20	N2,P6
Potassium	mg/kg	2220	894	907	4140	3390	215	129	75-125	20	20	M3
Sodium	mg/kg	ND	894	907	936	767	95	75	75-125	20	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729370
QC Batch Method: EPA 3010
Analysis Method: EPA 6010
Analysis Description: 6010 MET TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3347552 Matrix: Water

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	04/24/23 22:19	
Barium	mg/L	<0.50	0.50	04/24/23 22:19	
Cadmium	mg/L	<0.0050	0.0050	04/24/23 22:19	
Chromium	mg/L	<0.010	0.010	04/24/23 22:19	
Lead	mg/L	<0.010	0.010	04/24/23 22:19	
Selenium	mg/L	<0.010	0.010	04/24/23 22:19	
Silver	mg/L	<0.010	0.010	04/24/23 22:19	
Zinc	mg/L	<0.020	0.020	04/24/23 22:19	

LABORATORY CONTROL SAMPLE: 3347553

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	0.96	96	80-120	
Barium	mg/L	1	0.94	94	80-120	
Cadmium	mg/L	1	0.92	92	80-120	
Chromium	mg/L	1	0.93	93	80-120	
Lead	mg/L	1	0.88	88	80-120	
Selenium	mg/L	1	0.95	95	80-120	
Silver	mg/L	0.5	0.45	90	80-120	
Zinc	mg/L	1	0.92	92	80-120	

MATRIX SPIKE SAMPLE: 3347554

Parameter	Units	50341662001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10	100	50-150	
Barium	mg/L	<5.0	10	11.0	96	50-150	
Cadmium	mg/L	<0.050	10	9.5	95	50-150	
Chromium	mg/L	0.13	10	9.8	97	50-150	
Lead	mg/L	<0.10	10	9.1	90	50-150	
Selenium	mg/L	<0.10	10	9.8	98	50-150	
Silver	mg/L	<0.10	5	4.7	94	50-150	
Zinc	mg/L	3.2	10	12.6	95	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

MATRIX SPIKE SAMPLE: 3347555

Parameter	Units	50341914001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	11.0	109	50-150	
Barium	mg/L	ND	10	9.7	97	50-150	
Cadmium	mg/L	ND	10	10.1	101	50-150	
Chromium	mg/L	ND	10	9.2	91	50-150	
Lead	mg/L	ND	10	8.2	82	50-150	
Selenium	mg/L	ND	10	10.4	104	50-150	
Silver	mg/L	ND	5	5.0	99	50-150	
Zinc	mg/L	ND	10	10.0	96	50-150	

MATRIX SPIKE SAMPLE: 3347556

Parameter	Units	50342005001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	12.5	97	50-150	
Cadmium	mg/L	ND	10	9.5	95	50-150	
Chromium	mg/L	ND	10	9.4	94	50-150	
Lead	mg/L	ND	10	8.9	89	50-150	
Selenium	mg/L	ND	10	9.9	99	50-150	
Silver	mg/L	ND	5	4.7	93	50-150	
Zinc	mg/L	2360 ug/L	10	11.6	93	50-150	

MATRIX SPIKE SAMPLE: 3347557

Parameter	Units	50342006001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.3	93	50-150	
Barium	mg/L	ND	10	9.8	97	50-150	
Cadmium	mg/L	ND	10	9.6	96	50-150	
Chromium	mg/L	18.2	10	27.6	94	50-150	
Lead	mg/L	0.17	10	8.2	80	50-150	
Selenium	mg/L	ND	10	7.4	74	50-150	
Silver	mg/L	ND	5	4.9	97	50-150	
Zinc	mg/L	2750 ug/L	10	11.4	86	50-150	

MATRIX SPIKE SAMPLE: 3347558

Parameter	Units	50342062001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.3	103	50-150	
Barium	mg/L	ND	10	9.6	95	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.7	97	50-150	
Lead	mg/L	ND	10	9.0	90	50-150	
Selenium	mg/L	ND	10	10.0	100	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

MATRIX SPIKE SAMPLE: 3347558

Parameter	Units	50342062001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Silver	mg/L	ND	5	4.8	95	50-150	
Zinc	mg/L	ND	10	9.7	96	50-150	

MATRIX SPIKE SAMPLE: 3347559

Parameter	Units	50342150001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.2	102	50-150	
Barium	mg/L	ND	10	9.9	98	50-150	
Cadmium	mg/L	ND	10	9.6	95	50-150	
Chromium	mg/L	0.12	10	9.7	96	50-150	
Lead	mg/L	ND	10	8.9	89	50-150	
Selenium	mg/L	ND	10	9.9	99	50-150	
Silver	mg/L	ND	5	4.7	95	50-150	
Zinc	mg/L	2400 ug/L	10	11.9	95	50-150	

MATRIX SPIKE SAMPLE: 3347560

Parameter	Units	50342437001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	10.0	96	50-150	
Cadmium	mg/L	ND	10	9.5	95	50-150	
Chromium	mg/L	ND	10	9.5	95	50-150	
Lead	mg/L	ND	10	8.8	88	50-150	
Selenium	mg/L	ND	10	9.9	99	50-150	
Silver	mg/L	ND	5	4.6	93	50-150	
Zinc	mg/L	ND	10	9.5	94	50-150	

MATRIX SPIKE SAMPLE: 3347561

Parameter	Units	50342113001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.3	103	50-150	
Barium	mg/L	ND	10	10	96	50-150	
Cadmium	mg/L	ND	10	9.6	96	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	ND	10	8.8	88	50-150	
Selenium	mg/L	ND	10	10.0	100	50-150	
Silver	mg/L	ND	5	4.7	94	50-150	
Zinc	mg/L	ND	10	9.5	95	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

MATRIX SPIKE SAMPLE: 3347562

Parameter	Units	50342119001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.0	100	50-150	
Barium	mg/L	<5.0	10	9.6	95	50-150	
Cadmium	mg/L	<0.050	10	9.4	94	50-150	
Chromium	mg/L	<0.10	10	9.5	95	50-150	
Lead	mg/L	<0.10	10	8.8	88	50-150	
Selenium	mg/L	<0.10	10	9.8	98	50-150	
Silver	mg/L	<0.10	5	4.6	93	50-150	
Zinc	mg/L	3.0	10	12.3	93	50-150	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3347563 3347564

Parameter	Units	50342146001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	10.4	10.3	104	102	50-150	2	20	
Barium	mg/L	ND	10	10	11.5	11.4	98	97	50-150	1	20	
Cadmium	mg/L	ND	10	10	9.7	9.5	97	95	50-150	2	20	
Chromium	mg/L	ND	10	10	9.8	9.5	97	95	50-150	3	20	
Lead	mg/L	ND	10	10	9.0	8.9	90	89	50-150	2	20	
Selenium	mg/L	ND	10	10	10.1	9.9	101	99	50-150	2	20	
Silver	mg/L	ND	5	5	4.8	4.7	95	93	50-150	2	20	
Zinc	mg/L	ND	10	10	9.7	9.4	97	94	50-150	3	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729422
QC Batch Method: EPA 3050B
Analysis Method: EPA 6020
Analysis Description: 6020 MET
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50342119001

METHOD BLANK: 3347779 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	04/26/23 19:59	
Cadmium	mg/kg	<0.050	0.050	04/26/23 19:59	
Chromium	mg/kg	<0.20	0.20	04/26/23 19:59	
Copper	mg/kg	<0.10	0.10	04/26/23 19:59	
Lead	mg/kg	<0.10	0.10	04/26/23 19:59	
Molybdenum	mg/kg	<0.10	0.10	04/26/23 19:59	N2
Nickel	mg/kg	<0.050	0.050	04/26/23 19:59	
Selenium	mg/kg	<0.10	0.10	04/26/23 19:59	
Silver	mg/kg	<0.050	0.050	04/26/23 19:59	
Zinc	mg/kg	<0.50	0.50	04/26/23 19:59	

LABORATORY CONTROL SAMPLE: 3347780

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.8	94	80-120	
Cadmium	mg/kg	4	3.9	98	80-120	
Chromium	mg/kg	4	4.1	102	80-120	
Copper	mg/kg	4	4.0	99	80-120	
Lead	mg/kg	4	4.0	99	80-120	
Molybdenum	mg/kg	4	4.0	99	80-120	N2
Nickel	mg/kg	4	3.9	97	80-120	
Selenium	mg/kg	4	3.8	95	80-120	
Silver	mg/kg	4	4.0	100	80-120	
Zinc	mg/kg	4	3.9	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3347781 3347782

Parameter	Units	50342755007 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	<1930 ug/kg	3.9	3.8	3.9	3.8	87	87	75-125	3	20	
Cadmium	mg/kg	<193 ug/kg	3.9	3.8	3.8	3.7	95	97	75-125	1	20	
Chromium	mg/kg	2430 ug/kg	3.9	3.8	6.5	7.0	106	122	75-125	7	20	
Copper	mg/kg	1350 ug/kg	3.9	3.8	4.8	4.7	87	88	75-125	2	20	
Lead	mg/kg	<9650 ug/kg	3.9	3.8	5.0	5.1	101	106	75-125	2	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3347781 3347782

Parameter	Units	50342755007 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Molybdenum	mg/kg	<965 ug/kg	3.9	3.8	3.4	3.2	86	85	75-125	4	20	N2
Nickel	mg/kg	1870 ug/kg	3.9	3.8	5.2	5.5	86	96	75-125	5	20	
Selenium	mg/kg	967 ug/kg	3.9	3.8	4.1	4.3	82	87	75-125	3	20	
Silver	mg/kg	<96.5 ug/kg	3.9	3.8	3.7	3.7	96	98	75-125	1	20	
Zinc	mg/kg	3670 ug/kg	3.9	3.8	7.3	9.2	94	147	75-125	23	20	M0,R1

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729060
QC Batch Method: EPA 8260
Analysis Method: EPA 8260
Analysis Description: 8260 MSV 5030 Low
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3345432
Matrix: Solid

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	04/21/23 01:09	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
2-Butanone (MEK)	mg/kg	<0.025	0.025	04/21/23 01:09	
2-Hexanone	mg/kg	<0.10	0.10	04/21/23 01:09	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	04/21/23 01:09	
Acetone	mg/kg	<0.10	0.10	04/21/23 01:09	
Acrolein	mg/kg	<0.10	0.10	04/21/23 01:09	
Acrylonitrile	mg/kg	<0.10	0.10	04/21/23 01:09	
Benzene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Bromodichloromethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Bromoform	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Bromomethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Carbon disulfide	mg/kg	<0.010	0.010	04/21/23 01:09	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Chlorobenzene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Chloroethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Chloroform	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Chloromethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Dibromochloromethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Dibromomethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Ethyl methacrylate	mg/kg	<0.10	0.10	04/21/23 01:09	
Ethylbenzene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Iodomethane	mg/kg	<0.10	0.10	04/21/23 01:09	
Methyl methacrylate	mg/kg	<0.10	0.10	04/21/23 01:09	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	04/21/23 01:09	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

METHOD BLANK: 3345432 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	04/21/23 01:09	
Styrene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Tetrachloroethene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Toluene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	04/21/23 01:09	
Trichloroethene	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Vinyl acetate	mg/kg	<0.10	0.10	04/21/23 01:09	
Vinyl chloride	mg/kg	<0.0050	0.0050	04/21/23 01:09	
Xylene (Total)	mg/kg	<0.010	0.010	04/21/23 01:09	
4-Bromofluorobenzene (S)	%.	101	63-132	04/21/23 01:09	
Dibromofluoromethane (S)	%.	101	75-135	04/21/23 01:09	1d
Toluene-d8 (S)	%.	100	65-148	04/21/23 01:09	

LABORATORY CONTROL SAMPLE: 3345433

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	0.05	0.044	87	74-134	
1,1,1-Trichloroethane	mg/kg	0.05	0.042	85	66-133	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.040	80	62-131	
1,1,2-Trichloroethane	mg/kg	0.05	0.046	91	73-128	
1,1-Dichloroethane	mg/kg	0.05	0.043	85	67-128	
1,1-Dichloroethene	mg/kg	0.05	0.040	81	65-138	
1,2,3-Trichloropropane	mg/kg	0.05	0.044	89	69-128	
1,2-Dibromo-3-chloropropane	mg/kg	0.05	0.047	94	62-133	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.045	90	74-131	
1,2-Dichlorobenzene	mg/kg	0.05	0.039	77	68-130	
1,2-Dichloroethane	mg/kg	0.05	0.044	88	62-129	
1,2-Dichloropropane	mg/kg	0.05	0.043	86	64-132	
1,3-Dichlorobenzene	mg/kg	0.05	0.037	74	66-131	
1,4-Dichlorobenzene	mg/kg	0.05	0.037	74	66-129	
2-Butanone (MEK)	mg/kg	0.25	0.22	86	57-139	
2-Hexanone	mg/kg	0.25	0.23	91	59-130	
4-Methyl-2-pentanone (MIBK)	mg/kg	0.25	0.23	90	62-132	
Acetone	mg/kg	0.25	0.19	77	45-139	
Acrolein	mg/kg	1	0.70	70	59-123	
Acrylonitrile	mg/kg	0.25	0.21	84	64-135	
Benzene	mg/kg	0.05	0.041	82	65-128	
Bromodichloromethane	mg/kg	0.05	0.047	93	69-132	
Bromoform	mg/kg	0.05	0.046	93	69-140	
Bromomethane	mg/kg	0.05	0.036	71	42-162	
Carbon disulfide	mg/kg	0.05	0.041	82	64-132	

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REPORT OF LABORATORY ANALYSIS

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Grand Rapids, MI 49512
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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

LABORATORY CONTROL SAMPLE: 3345433

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Carbon tetrachloride	mg/kg	0.05	0.040	80	65-141	
Chlorobenzene	mg/kg	0.05	0.039	78	69-124	
Chloroethane	mg/kg	0.05	0.030	61	64-136	L2
Chloroform	mg/kg	0.05	0.044	87	67-122	
Chloromethane	mg/kg	0.05	0.038	75	49-130	
cis-1,2-Dichloroethene	mg/kg	0.05	0.043	85	64-131	
cis-1,3-Dichloropropene	mg/kg	0.05	0.045	90	72-134	
Dibromochloromethane	mg/kg	0.05	0.047	94	73-136	
Dibromomethane	mg/kg	0.05	0.046	91	70-131	
Dichlorodifluoromethane	mg/kg	0.05	0.030	59	10-117	
Ethyl methacrylate	mg/kg	0.05	<0.10	95	70-132	
Ethylbenzene	mg/kg	0.05	0.039	78	67-127	
Iodomethane	mg/kg	0.05	<0.10	85	42-150	
Methyl methacrylate	mg/kg	0.05	<0.10	95	62-140	
Methyl-tert-butyl ether	mg/kg	0.05	0.049	97	66-135	
Methylene Chloride	mg/kg	0.05	0.043	87	66-137	
Styrene	mg/kg	0.05	0.040	80	72-129	
Tetrachloroethene	mg/kg	0.05	0.036	72	62-135	
Toluene	mg/kg	0.05	0.039	78	65-123	
trans-1,2-Dichloroethene	mg/kg	0.05	0.041	81	63-131	
trans-1,3-Dichloropropene	mg/kg	0.05	0.046	92	71-135	
trans-1,4-Dichloro-2-butene	mg/kg	0.05	<0.10	81	62-156	
Trichloroethene	mg/kg	0.05	0.041	82	64-135	
Trichlorofluoromethane	mg/kg	0.05	0.035	71	56-142	
Vinyl acetate	mg/kg	0.2	0.12	58	47-142	
Vinyl chloride	mg/kg	0.05	0.035	70	54-134	
Xylene (Total)	mg/kg	0.15	0.12	77	65-124	
4-Bromofluorobenzene (S)	%.			102	63-132	
Dibromofluoromethane (S)	%.			102	75-135	
Toluene-d8 (S)	%.			100	65-148	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 730196
QC Batch Method: EPA 5030/8260
Analysis Method: EPA 5030/8260
Analysis Description: 8260 MSV TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3351032 Matrix: Water

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	04/27/23 10:20	
1,2-Dichloroethane	mg/L	<0.050	0.050	04/27/23 10:20	
2-Butanone (MEK)	mg/L	<1.0	1.0	04/27/23 10:20	
Benzene	mg/L	<0.050	0.050	04/27/23 10:20	
Carbon tetrachloride	mg/L	<0.050	0.050	04/27/23 10:20	
Chlorobenzene	mg/L	<0.050	0.050	04/27/23 10:20	
Chloroform	mg/L	<0.050	0.050	04/27/23 10:20	
Tetrachloroethene	mg/L	<0.050	0.050	04/27/23 10:20	
Trichloroethene	mg/L	<0.050	0.050	04/27/23 10:20	
Vinyl chloride	mg/L	<0.020	0.020	04/27/23 10:20	
4-Bromofluorobenzene (S)	%.	99	79-124	04/27/23 10:20	
Dibromofluoromethane (S)	%.	100	82-128	04/27/23 10:20	
Toluene-d8 (S)	%.	98	73-122	04/27/23 10:20	

LABORATORY CONTROL SAMPLE: 3351033

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.56	112	73-133	
1,2-Dichloroethane	mg/L	0.5	0.53	106	70-124	
2-Butanone (MEK)	mg/L	2.5	2.6	102	59-134	
Benzene	mg/L	0.5	0.53	106	74-124	
Carbon tetrachloride	mg/L	0.5	0.53	106	78-132	
Chlorobenzene	mg/L	0.5	0.53	106	77-121	
Chloroform	mg/L	0.5	0.55	110	75-118	
Tetrachloroethene	mg/L	0.5	0.53	107	73-132	
Trichloroethene	mg/L	0.5	0.56	111	75-127	
Vinyl chloride	mg/L	0.5	0.47	94	48-133	
4-Bromofluorobenzene (S)	%.			102	79-124	
Dibromofluoromethane (S)	%.			101	82-128	
Toluene-d8 (S)	%.			98	73-122	

MATRIX SPIKE SAMPLE: 3351034

Parameter	Units	50342273001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.53	107	65-139	
1,2-Dichloroethane	mg/L	ND	0.5	0.53	106	55-146	
2-Butanone (MEK)	mg/L	ND	2.5	2.4	96	45-155	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

MATRIX SPIKE SAMPLE: 3351034

Parameter	Units	50342273001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	ND	0.5	0.53	106	65-137	
Carbon tetrachloride	mg/L	ND	0.5	0.51	102	65-156	
Chlorobenzene	mg/L	ND	0.5	0.53	105	54-135	
Chloroform	mg/L	ND	0.5	0.54	108	64-133	
Tetrachloroethene	mg/L	ND	0.5	0.53	105	43-149	
Trichloroethene	mg/L	ND	0.5	0.54	109	52-145	
Vinyl chloride	mg/L	ND	0.5	0.46	92	43-139	
4-Bromofluorobenzene (S)	%.				105	79-124	
Dibromofluoromethane (S)	%.				98	82-128	
Toluene-d8 (S)	%.				98	73-122	

MATRIX SPIKE SAMPLE: 3351035

Parameter	Units	50342345005 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.55	110	65-139	
1,2-Dichloroethane	mg/L	ND	0.5	0.53	106	55-146	
2-Butanone (MEK)	mg/L	ND	2.5	2.4	96	45-155	
Benzene	mg/L	ND	0.5	0.54	107	65-137	
Carbon tetrachloride	mg/L	ND	0.5	0.53	106	65-156	
Chlorobenzene	mg/L	ND	0.5	0.50	101	54-135	
Chloroform	mg/L	ND	0.5	0.55	110	64-133	
Tetrachloroethene	mg/L	933 ug/L	0.5	1.4	95	43-149	
Trichloroethene	mg/L	ND	0.5	0.55	110	52-145	
Vinyl chloride	mg/L	ND	0.5	0.48	95	43-139	
4-Bromofluorobenzene (S)	%.				103	79-124	
Dibromofluoromethane (S)	%.				98	82-128	
Toluene-d8 (S)	%.				98	73-122	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 728301
QC Batch Method: EPA 3546
Analysis Method: EPA 8081
Analysis Description: 8081 GCS Pesticides
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3342464 Matrix: Solid

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	04/18/23 00:47	
4,4'-DDE	ug/kg	<5.0	5.0	04/18/23 00:47	
4,4'-DDT	ug/kg	<5.0	5.0	04/18/23 00:47	
Aldrin	ug/kg	<2.5	2.5	04/18/23 00:47	
alpha-BHC	ug/kg	<2.5	2.5	04/18/23 00:47	
alpha-Chlordane	ug/kg	<2.5	2.5	04/18/23 00:47	
beta-BHC	ug/kg	<2.5	2.5	04/18/23 00:47	
Chlordane (Technical)	ug/kg	<50.0	50.0	04/18/23 00:47	
delta-BHC	ug/kg	<2.5	2.5	04/18/23 00:47	
Dieldrin	ug/kg	<5.0	5.0	04/18/23 00:47	
Endosulfan I	ug/kg	<2.5	2.5	04/18/23 00:47	
Endosulfan II	ug/kg	<5.0	5.0	04/18/23 00:47	
Endosulfan sulfate	ug/kg	<5.0	5.0	04/18/23 00:47	
Endrin	ug/kg	<5.0	5.0	04/18/23 00:47	
Endrin aldehyde	ug/kg	<5.0	5.0	04/18/23 00:47	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	04/18/23 00:47	
gamma-Chlordane	ug/kg	<2.5	2.5	04/18/23 00:47	
Heptachlor	ug/kg	<2.5	2.5	04/18/23 00:47	
Heptachlor epoxide	ug/kg	<2.5	2.5	04/18/23 00:47	
Toxaphene	ug/kg	<50.0	50.0	04/18/23 00:47	
Decachlorobiphenyl (S)	%.	84	10-147	04/18/23 00:47	

LABORATORY CONTROL SAMPLE: 3342465

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	20	17.0	85	40-148	
4,4'-DDE	ug/kg	20	17.4	87	42-145	
4,4'-DDT	ug/kg	20	16.4	82	28-169	
Aldrin	ug/kg	10	8.5	85	42-137	
alpha-BHC	ug/kg	10	8.1	81	40-141	
alpha-Chlordane	ug/kg	10	8.5	85	38-141	
beta-BHC	ug/kg	10	8.3	83	42-137	
delta-BHC	ug/kg	10	7.0	70	33-132	
Dieldrin	ug/kg	20	17.7	89	42-143	
Endosulfan I	ug/kg	10	8.1	81	38-142	
Endosulfan II	ug/kg	20	17.4	87	42-153	
Endosulfan sulfate	ug/kg	20	17.3	86	36-133	
Endrin	ug/kg	20	17.1	86	43-145	
Endrin aldehyde	ug/kg	20	19.5	98	40-138	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

LABORATORY CONTROL SAMPLE: 3342465

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	10	8.6	86	42-142	
gamma-Chlordane	ug/kg	10	8.0	80	44-148	
Heptachlor	ug/kg	10	8.2	82	42-143	
Heptachlor epoxide	ug/kg	10	8.7	87	41-136	
Decachlorobiphenyl (S)	%.			93	10-147	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3342466 3342467

Parameter	Units	60425917001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	20.9	21.2	11.9	12.0	57	56	10-171	0	20	
4,4'-DDE	ug/kg	ND	20.9	21.2	14.5	13.2	59	52	10-146	10	20	
4,4'-DDT	ug/kg	ND	20.9	21.2	11.4	11.1	54	52	10-165	3	20	
Aldrin	ug/kg	ND	10.5	10.6	6.6	6.3	63	60	12-141	4	20	
alpha-BHC	ug/kg	ND	10.5	10.6	6.5	6.5	62	61	14-157	0	20	
alpha-Chlordane	ug/kg	ND	10.5	10.6	6.2	6.1	59	57	10-153	3	20	
beta-BHC	ug/kg	ND	10.5	10.6	6.9	6.5	66	61	10-163	6	20	
delta-BHC	ug/kg	ND	10.5	10.6	5.6	5.3	53	50	10-155	6	20	
Dieldrin	ug/kg	ND	20.9	21.2	12.8	12.7	61	60	14-141	0	20	
Endosulfan I	ug/kg	ND	10.5	10.6	5.9	5.9	56	55	10-141	0	20	
Endosulfan II	ug/kg	ND	20.9	21.2	11.7	11.8	56	55	10-149	0	20	
Endosulfan sulfate	ug/kg	ND	20.9	21.2	11.6	11.5	55	54	10-126	1	20	
Endrin	ug/kg	ND	20.9	21.2	12.3	12.3	59	58	13-136	0	20	
Endrin aldehyde	ug/kg	ND	20.9	21.2	12.8	13.2	61	62	10-150	3	20	
gamma-BHC (Lindane)	ug/kg	ND	10.5	10.6	6.9	6.8	66	64	18-152	2	20	
gamma-Chlordane	ug/kg	ND	10.5	10.6	6.2	6.0	59	56	10-159	3	20	
Heptachlor	ug/kg	ND	10.5	10.6	6.5	6.2	62	59	10-162	4	20	
Heptachlor epoxide	ug/kg	ND	10.5	10.6	6.5	6.4	62	60	15-142	1	20	
Decachlorobiphenyl (S)	%.						51	50	10-147			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729630
QC Batch Method: EPA 3510
Analysis Method: EPA 8081
Analysis Description: 8081 GCS TCLP Pesticides
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3348373 Matrix: Water

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.0010	0.0010	04/25/23 18:33	
Endrin	mg/L	<0.00010	0.00010	04/25/23 18:33	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	04/25/23 18:33	
Heptachlor	mg/L	<0.000050	0.000050	04/25/23 18:33	
Heptachlor epoxide	mg/L	<0.000050	0.000050	04/25/23 18:33	
Methoxychlor	mg/L	<0.00050	0.00050	04/25/23 18:33	
Toxaphene	mg/L	<0.0010	0.0010	04/25/23 18:33	
Decachlorobiphenyl (S)	%.	93	10-124	04/25/23 18:33	

LABORATORY CONTROL SAMPLE: 3348374

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0015	75	29-177	
gamma-BHC (Lindane)	mg/L	0.001	0.00077	77	26-163	
Heptachlor	mg/L	0.001	0.00076	76	12-157	
Heptachlor epoxide	mg/L	0.001	0.00078	78	27-168	
Methoxychlor	mg/L	0.01	0.0091	91	28-174	
Decachlorobiphenyl (S)	%.			81	10-124	

MATRIX SPIKE SAMPLE: 3348375

Parameter	Units	50342119001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	<0.00050	0.01	0.0031	31	10-179	
gamma-BHC (Lindane)	mg/L	<0.00025	0.005	0.0021	42	11-157	
Heptachlor	mg/L	<0.00025	0.005	0.0017	34	10-154	
Heptachlor epoxide	mg/L	<0.00025	0.005	0.0016	33	10-170	
Methoxychlor	mg/L	<0.0025	0.05	0.017	34	10-162	
Decachlorobiphenyl (S)	%.				25	10-124	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 728678 Analysis Method: EPA 8082
QC Batch Method: EPA 3546 Analysis Description: 8082 PCB Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3343889 Matrix: Solid

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<100	100	04/24/23 19:58	
PCB-1221 (Aroclor 1221)	ug/kg	<100	100	04/24/23 19:58	
PCB-1232 (Aroclor 1232)	ug/kg	<100	100	04/24/23 19:58	
PCB-1242 (Aroclor 1242)	ug/kg	<100	100	04/24/23 19:58	
PCB-1248 (Aroclor 1248)	ug/kg	<100	100	04/24/23 19:58	
PCB-1254 (Aroclor 1254)	ug/kg	<100	100	04/24/23 19:58	
PCB-1260 (Aroclor 1260)	ug/kg	<100	100	04/24/23 19:58	
Tetrachloro-m-xylene (S)	%.	96	10-133	04/24/23 19:58	

LABORATORY CONTROL SAMPLE: 3343890

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	333	280	84	50-120	
PCB-1260 (Aroclor 1260)	ug/kg	333	347	104	40-122	
Tetrachloro-m-xylene (S)	%.			83	10-133	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3343891 3343892

Parameter	Units	50342121008 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	413	413	293	301	71	73	10-154	2	20	
PCB-1260 (Aroclor 1260)	ug/kg	ND	413	413	218	230	53	56	10-165	5	20	
Tetrachloro-m-xylene (S)	%.						51	52	10-133			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729642
QC Batch Method: EPA 8151
Analysis Method: EPA 8151A
Analysis Description: 8151 GCS TCLP Herbicides
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3348415 Matrix: Water

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	04/27/23 19:00	
2,4-D	mg/L	<0.0010	0.0010	04/27/23 19:00	
2,4-DCAA (S)	%.	82	22-132	04/27/23 19:18	

LABORATORY CONTROL SAMPLE: 3348416

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0036	72	37-122	
2,4-D	mg/L	0.005	0.0035	70	24-129	
2,4-DCAA (S)	%.			87	22-132	

MATRIX SPIKE SAMPLE: 3348417

Parameter	Units	50342119001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0050	0.025	0.019	77	20-155	
2,4-D	mg/L	<0.0050	0.025	0.021	83	12-159	
2,4-DCAA (S)	%.				71	22-132	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch:	728740	Analysis Method:	EPA 8270
QC Batch Method:	EPA 3546	Analysis Description:	8270 Solid MSSV Microwave Short Spike
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3344100 Matrix: Solid

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<330	330	04/19/23 17:22	
2,4,5-Trichlorophenol	ug/kg	<330	330	04/19/23 17:22	
2,4,6-Trichlorophenol	ug/kg	<330	330	04/19/23 17:22	
2,4-Dichlorophenol	ug/kg	<330	330	04/19/23 17:22	
2,4-Dimethylphenol	ug/kg	<330	330	04/19/23 17:22	
2,4-Dinitrophenol	ug/kg	<1600	1600	04/19/23 17:22	
2,4-Dinitrotoluene	ug/kg	<330	330	04/19/23 17:22	
2,6-Dinitrotoluene	ug/kg	<330	330	04/19/23 17:22	
2-Chloronaphthalene	ug/kg	<330	330	04/19/23 17:22	
2-Chlorophenol	ug/kg	<330	330	04/19/23 17:22	
2-Methylnaphthalene	ug/kg	<330	330	04/19/23 17:22	
2-Methylphenol(o-Cresol)	ug/kg	<330	330	04/19/23 17:22	
2-Nitroaniline	ug/kg	<330	330	04/19/23 17:22	
2-Nitrophenol	ug/kg	<330	330	04/19/23 17:22	
3&4-Methylphenol(m&p Cresol)	ug/kg	<660	660	04/19/23 17:22	
3,3'-Dichlorobenzidine	ug/kg	<660	660	04/19/23 17:22	
3-Nitroaniline	ug/kg	<330	330	04/19/23 17:22	
4,6-Dinitro-2-methylphenol	ug/kg	<660	660	04/19/23 17:22	
4-Bromophenylphenyl ether	ug/kg	<330	330	04/19/23 17:22	
4-Chloro-3-methylphenol	ug/kg	<660	660	04/19/23 17:22	
4-Chloroaniline	ug/kg	<660	660	04/19/23 17:22	
4-Chlorophenylphenyl ether	ug/kg	<330	330	04/19/23 17:22	
4-Nitroaniline	ug/kg	<330	330	04/19/23 17:22	
4-Nitrophenol	ug/kg	<1600	1600	04/19/23 17:22	
Acenaphthene	ug/kg	<330	330	04/19/23 17:22	
Acenaphthylene	ug/kg	<330	330	04/19/23 17:22	
Anthracene	ug/kg	<330	330	04/19/23 17:22	
Benzo(a)anthracene	ug/kg	<330	330	04/19/23 17:22	
Benzo(a)pyrene	ug/kg	<330	330	04/19/23 17:22	
Benzo(b)fluoranthene	ug/kg	<330	330	04/19/23 17:22	
Benzo(g,h,i)perylene	ug/kg	<330	330	04/19/23 17:22	
Benzo(k)fluoranthene	ug/kg	<330	330	04/19/23 17:22	
Benzyl alcohol	ug/kg	<660	660	04/19/23 17:22	
bis(2-Chloroethoxy)methane	ug/kg	<330	330	04/19/23 17:22	
bis(2-Chloroethyl) ether	ug/kg	<330	330	04/19/23 17:22	
bis(2-Ethylhexyl)phthalate	ug/kg	<330	330	04/19/23 17:22	
bis(2chloro1methylethyl) ether	ug/kg	<330	330	04/19/23 17:22	
Butylbenzylphthalate	ug/kg	<330	330	04/19/23 17:22	
Chrysene	ug/kg	<330	330	04/19/23 17:22	
Di-n-butylphthalate	ug/kg	<330	330	04/19/23 17:22	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

METHOD BLANK: 3344100 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<330	330	04/19/23 17:22	
Dibenz(a,h)anthracene	ug/kg	<330	330	04/19/23 17:22	
Dibenzofuran	ug/kg	<330	330	04/19/23 17:22	
Diethylphthalate	ug/kg	<330	330	04/19/23 17:22	
Dimethylphthalate	ug/kg	<330	330	04/19/23 17:22	
Fluoranthene	ug/kg	<330	330	04/19/23 17:22	
Fluorene	ug/kg	<330	330	04/19/23 17:22	
Hexachloro-1,3-butadiene	ug/kg	<330	330	04/19/23 17:22	
Hexachlorobenzene	ug/kg	<330	330	04/19/23 17:22	
Hexachlorocyclopentadiene	ug/kg	<330	330	04/19/23 17:22	
Hexachloroethane	ug/kg	<330	330	04/19/23 17:22	
Indeno(1,2,3-cd)pyrene	ug/kg	<330	330	04/19/23 17:22	
Isophorone	ug/kg	<330	330	04/19/23 17:22	
N-Nitroso-di-n-propylamine	ug/kg	<330	330	04/19/23 17:22	
N-Nitrosodiphenylamine	ug/kg	<330	330	04/19/23 17:22	
Naphthalene	ug/kg	<330	330	04/19/23 17:22	
Nitrobenzene	ug/kg	<330	330	04/19/23 17:22	
Pentachlorophenol	ug/kg	<1600	1600	04/19/23 17:22	
Phenanthrene	ug/kg	<330	330	04/19/23 17:22	
Phenol	ug/kg	<330	330	04/19/23 17:22	
Pyrene	ug/kg	<330	330	04/19/23 17:22	
2,4,6-Tribromophenol (S)	%.	91	10-123	04/19/23 17:22	
2-Fluorobiphenyl (S)	%.	67	36-100	04/19/23 17:22	
2-Fluorophenol (S)	%.	69	22-114	04/19/23 17:22	
Nitrobenzene-d5 (S)	%.	54	35-110	04/19/23 17:22	
p-Terphenyl-d14 (S)	%.	79	29-117	04/19/23 17:22	
Phenol-d5 (S)	%.	70	35-115	04/19/23 17:22	

LABORATORY CONTROL SAMPLE: 3344101

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1670	1350	81	57-115	
2-Chlorophenol	ug/kg	1670	1360	82	55-103	
2-Methylnaphthalene	ug/kg	1670	1340	81	53-108	
4-Chloro-3-methylphenol	ug/kg	1670	1440	87	60-121	
4-Nitrophenol	ug/kg	1670	<1600	81	48-129	
Acenaphthene	ug/kg	1670	1320	79	57-102	
Acenaphthylene	ug/kg	1670	1440	86	56-103	
Anthracene	ug/kg	1670	1430	86	62-106	
Benzo(a)anthracene	ug/kg	1670	1420	85	63-110	
Benzo(a)pyrene	ug/kg	1670	1380	83	60-114	
Benzo(b)fluoranthene	ug/kg	1670	1320	79	61-119	
Benzo(g,h,i)perylene	ug/kg	1670	1230	74	62-109	
Benzo(k)fluoranthene	ug/kg	1670	1410	85	59-115	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

LABORATORY CONTROL SAMPLE: 3344101

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1670	1430	86	61-109	
Dibenz(a,h)anthracene	ug/kg	1670	1300	78	62-111	
Fluoranthene	ug/kg	1670	1440	87	65-113	
Fluorene	ug/kg	1670	1430	86	60-109	
Indeno(1,2,3-cd)pyrene	ug/kg	1670	1370	82	62-111	
N-Nitroso-di-n-propylamine	ug/kg	1670	1070	64	51-105	
Naphthalene	ug/kg	1670	1240	74	53-103	
Pentachlorophenol	ug/kg	1670	<1600	71	33-128	
Phenanthrene	ug/kg	1670	1460	87	62-108	
Phenol	ug/kg	1670	1270	76	45-112	
Pyrene	ug/kg	1670	1510	91	61-113	
2,4,6-Tribromophenol (S)	%.			100	10-123	
2-Fluorobiphenyl (S)	%.			68	36-100	
2-Fluorophenol (S)	%.			75	22-114	
Nitrobenzene-d5 (S)	%.			60	35-110	
p-Terphenyl-d14 (S)	%.			85	29-117	
Phenol-d5 (S)	%.			75	35-115	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch:	730010	Analysis Method:	EPA 8270
QC Batch Method:	EPA 3510	Analysis Description:	8270 TCLP MSSV
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3350168 Matrix: Water

Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	04/27/23 19:30	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	04/27/23 19:30	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	04/27/23 19:30	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	04/27/23 19:30	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	04/27/23 19:30	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	04/27/23 19:30	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	04/27/23 19:30	
Hexachlorobenzene	mg/L	<0.010	0.010	04/27/23 19:30	
Hexachloroethane	mg/L	<0.010	0.010	04/27/23 19:30	
Nitrobenzene	mg/L	<0.010	0.010	04/27/23 19:30	
Pentachlorophenol	mg/L	<0.050	0.050	04/27/23 19:30	
Pyridine	mg/L	<0.010	0.010	04/27/23 19:30	
2,4,6-Tribromophenol (S)	%.	124	37-127	04/27/23 19:30	
2-Fluorobiphenyl (S)	%.	69	36-92	04/27/23 19:30	
2-Fluorophenol (S)	%.	52	23-69	04/27/23 19:30	
Nitrobenzene-d5 (S)	%.	70	44-107	04/27/23 19:30	
p-Terphenyl-d14 (S)	%.	120	57-146	04/27/23 19:30	
Phenol-d5 (S)	%.	32	15-47	04/27/23 19:30	

LABORATORY CONTROL SAMPLE: 3350169

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.028	57	23-87	
2,4,5-Trichlorophenol	mg/L	0.05	0.052	104	43-131	
2,4,6-Trichlorophenol	mg/L	0.05	0.046	92	42-128	
2,4-Dinitrotoluene	mg/L	0.05	0.052	104	47-116	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.033	66	35-98	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.064	64	30-91	
Hexachloro-1,3-butadiene	mg/L	0.05	0.025	50	10-92	
Hexachlorobenzene	mg/L	0.05	0.036	71	34-89	
Hexachloroethane	mg/L	0.05	0.023	45	14-84	
Nitrobenzene	mg/L	0.05	0.036	72	39-113	
Pentachlorophenol	mg/L	0.05	<0.050	90	16-133	
Pyridine	mg/L	0.05	0.014	27	10-61	
2,4,6-Tribromophenol (S)	%.			117	37-127	
2-Fluorobiphenyl (S)	%.			63	36-92	
2-Fluorophenol (S)	%.			47	23-69	
Nitrobenzene-d5 (S)	%.			67	44-107	
p-Terphenyl-d14 (S)	%.			109	57-146	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

LABORATORY CONTROL SAMPLE: 3350169

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			31	15-47	

MATRIX SPIKE SAMPLE: 3350170

Parameter	Units	50342680001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.33	66	18-91	
2,4,5-Trichlorophenol	mg/L	ND	0.5	0.56	113	51-131	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.49	97	50-124	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.51	103	41-118	
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.34	67	23-115	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.64	64	18-110	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.33	65	10-95	
Hexachlorobenzene	mg/L	ND	0.5	0.34	69	13-107	
Hexachloroethane	mg/L	ND	0.5	0.29	57	10-88	
Nitrobenzene	mg/L	ND	0.5	0.36	72	36-113	
Pentachlorophenol	mg/L	ND	0.5	<0.50	87	27-139	
Pyridine	mg/L	ND	0.5	0.20	40	11-66	
2,4,6-Tribromophenol (S)	%.				118	37-127	
2-Fluorobiphenyl (S)	%.				73	36-92	
2-Fluorophenol (S)	%.				47	23-69	
Nitrobenzene-d5 (S)	%.				68	44-107	
p-Terphenyl-d14 (S)	%.				101	57-146	
Phenol-d5 (S)	%.				29	15-47	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 728227
QC Batch Method: SM 2540G
Analysis Method: SM 2540G
Analysis Description: Dry Weight/Percent Moisture
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

SAMPLE DUPLICATE: 3342252

Parameter	Units	50342119001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	97.1	97.1	0	5	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 728224 | Analysis Method: SM 2540G
QC Batch Method: SM 2540G | Analysis Description: 2540G Total Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

SAMPLE DUPLICATE: 3342250

Parameter	Units	50342119001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	2.9	3.0	1	10	N2

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch:	728229	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	2540G Total Volatile Solids
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

SAMPLE DUPLICATE: 3342251

Parameter	Units	50342119001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	46.7	46.9	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 729922
QC Batch Method: EPA 9038
Analysis Method: EPA 9038
Analysis Description: 9038 S2ulfate Soil
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50342119001

METHOD BLANK: 3349900 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<100	100	04/26/23 11:12	N2

LABORATORY CONTROL SAMPLE: 3349901

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	201	188	94	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3349902 3349903

Parameter	Units	50342119001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	<3410	6850	6850	11500	11800	137	141	90-110	2	20	M3,N2, P4

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch:	728775	Analysis Method:	EPA 9045
QC Batch Method:	EPA 9045	Analysis Description:	9045 pH
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

SAMPLE DUPLICATE: 3344254

Parameter	Units	50342398001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.2	8.2	0	2	H3

SAMPLE DUPLICATE: 3344255

Parameter	Units	50342396001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	9.4	9.4	0	2	H3

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 728894
QC Batch Method: EPA 351.2
Analysis Method: EPA 351.2
Analysis Description: 351.2 TKN
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50342119001

METHOD BLANK: 3344777 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<48.5	48.5	04/21/23 13:44	N2

LABORATORY CONTROL SAMPLE: 3344778

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	485	455	94	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3344779 3344780

Parameter	Units	50342417001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	48100	2810	2730	52600	51700	158	130	90-110	2	20	M3,N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 730140
QC Batch Method: EPA 353.2
Analysis Method: EPA 353.2
Analysis Description: 353.2 Nitrate + Nitrite
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3350853 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	04/26/23 22:43	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	04/26/23 22:43	N2

LABORATORY CONTROL SAMPLE: 3350854

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	9.8	98	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.9	19.9	100	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3350855 3350856

Parameter	Units	50342562001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	239	142	142	370	369	92	91	90-110	0	20	N2
Nitrogen, NO2 plus NO3	mg/kg	320	283	283	596	596	97	97	90-110	0	20	N2

MATRIX SPIKE SAMPLE: 3350857

Parameter	Units	50342206002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	ND	10.8	15.1	101	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	ND	21.6	26.6	103	90-110	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch:	728196	Analysis Method:	SM 4500-NH3 G
QC Batch Method:	SM 4500-NH3 B	Analysis Description:	4500 Ammonia, Distilled
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3342167 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	04/17/23 14:26	N2

LABORATORY CONTROL SAMPLE: 3342168

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.8	20.7	105	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3342169 3342170

Parameter	Units	52114862001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	2900	185	178	3240	3130	184	128	90-110	3	20	N2,P6

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch: 730251
QC Batch Method: SM 4500-Cl-E
Analysis Method: SM 4500-Cl-E
Analysis Description: 4500 Chloride
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3351248 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.6	99.6	04/27/23 14:39	N2

LABORATORY CONTROL SAMPLE: 3351249

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	205	103	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3351250 3351251

Parameter	Units	50342119001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	4240	34100	34100	41300	40700	108	107	90-110	2	20	N2

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REPORT OF LABORATORY ANALYSIS

Pace Analytical®
www.pacelabs.com

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50342119

QC Batch:	730157	Analysis Method:	SM 4500-CN-E
QC Batch Method:	SM 4500-CN-C	Analysis Description:	4500CNE Cyanide
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50342119001

METHOD BLANK: 3350901 Matrix: Solid
Associated Lab Samples: 50342119001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Cyanide	mg/kg	<0.48	0.48	04/27/23 14:47	N2

LABORATORY CONTROL SAMPLE: 3350902

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/kg	4.8	4.8	100	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3350903 3350904

Parameter	Units	50342119001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/kg	<16.5	164	164	160	150	95	89	90-110	6	20	M0,N2

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50342119

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample volume.
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H3	Sample was received or analysis requested beyond the recognized method holding time.
L2	Analyte recovery in the laboratory control sample (LCS) was below QC limits. Results for this analyte in associated samples may be biased low.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P4	Sample field preservation does not meet EPA or method recommendations for this analysis.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids
Pace Project No.: 50342119

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50342119001	**Digestor #3**	EPA 3546	728301	EPA 8081	728371
50342119001	**Digestor #3**	EPA 3510	729630	EPA 8081	729726
50342119001	**Digestor #3**	EPA 3546	728678	EPA 8082	729570
50342119001	**Digestor #3**	EPA 8151	729642	EPA 8151A	730277
50342119001	**Digestor #3**	EPA 3050	729313	EPA 6010	729948
50342119001	**Digestor #3**	EPA 3010	729370	EPA 6010	729613
50342119001	**Digestor #3**	EPA 3050B	729422	EPA 6020	729925
50342119001	**Digestor #3**	EPA 7470	729481	EPA 7470	729767
50342119001	**Digestor #3**	EPA 7471	728535	EPA 7471	728897
50342119001	**Digestor #3**	EPA 3546	728740	EPA 8270	728826
50342119001	**Digestor #3**	EPA 3510	730010	EPA 8270	730489
50342119001	**Digestor #3**	EPA 8260	729060		
50342119001	**Digestor #3**	EPA 5030/8260	730196		
50342119001	**Digestor #3**	SM 2540G	728227		
50342119001	**Digestor #3**	SM 2540G	728224		
50342119001	**Digestor #3**	SM 2540G	728229		
50342119001	**Digestor #3**	EPA 9038	729922	EPA 9038	729928
50342119001	**Digestor #3**	EPA 9045	728775		
50342119001	**Digestor #3**	EPA 351.2	728894	EPA 351.2	729201
50342119001	**Digestor #3**	EPA 353.2	730140	EPA 353.2	730142
50342119001	**Digestor #3**	SM 4500-NH3 B	728196	SM 4500-NH3 G	728270
50342119001	**Digestor #3**	SM 4500-Cl-E	730251	SM 4500-Cl-E	730252
50342119001	**Digestor #3**	SM 4500-CN-C	730157	SM 4500-CN-E	730360

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CHAIN-OF-CUSTODY / Analytical Request D

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields mus

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace Terms and Conditions found at https://info.pac

WO#: 50342119

50342119

Section A
Required Client Information:

Company:	City of Cadillac
Address:	1121 Plett Rd
Cadillac,	MI 49601
Email:	ctomaszewski@cadillac-mi.net
Phone:	231-775-2368 Fax:
Requested Due Date:	

Section B
Required Project Information:

Report To:	Cindy Tomaszewski
Copy To:	
Purchase Order #:	
Project Name:	Biosolids
Project #:	

Section C
Invoice Information:

Attention:	
Company Name:	
Address:	
Pace Quote:	
Pace Project Manager:	brian.hall@pacelabs.com,
Pace Profile #:	8311

Regulatory Agency

State / Location: MI

SAMPLE ID
One Character per box.
(A-Z, 0-9 / , -)
Sample Ids must be unique

MATRIX	CODE
Drinking Water	DW
Water	WT
Waste Water	WW
Product	P
Soil/Solid	SL
Oil	OL
Wipe	WP
Air	AR
Other	OT
Tissue	TS

Requested Analysis Filtered (Y/N): TCLP METALS/SVOC/PEST N; SVOC,Pest/PCBs N; Metals/Nutrients N; VOC N

ITEM #	SAMPLE ID	MATRIX CODE (see valid codes to left)	SAMPLE TYPE (G=GRAB C=COMP)	COLLECTED START DATE	START TIME	END DATE	END TIME	SAMPLE TEMP AT COLLECTION	# OF CONTAINERS	Unpreserved	H2SO4	HNO3	HCl	NaOH	Na2S2O3	Methanol	Other	Analyses Test	TCLP METALS/SVOC/PEST	SVOC,Pest/PCBs	Metals/Nutrients	VOC	Residual Chlorine (Y/N)
1	~~AT~~ DIGESTOR #3	SL	G	—		4-13	1420		4	X									X	X	X	X	N
2	E																						
3	~~TB~~ DIGESTOR																						
4	"Sorry"																						
5																							
6																							
7																							
8																							
9																							
10																							
11																							
12																							

ADDITIONAL COMMENTS	RELINQUISHED BY / AFFILIATION	DATE	TIME	ACCEPTED BY / AFFILIATION	DATE	TIME	SAMPLE CONDITIONS
NOTICE - LOW SOLIDS %	CA Tomaszewski City of Cadillac	4-13-23	1500	Ship via UPS / Pace	4.14.23	1040	

SAMPLER NAME AND SIGNATURE

PRINT Name of SAMPLER: TAYSHA LACZKO & K. MOOMEY

SIGNATURE of SAMPLER: CA Tomaszewski for Samplers

DATE Signed: 04-13-23

TEMP in C	Received on Ice (Y/N)	Custody Sealed Cooler (Y/N)	Samples Intact (Y/N)

Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 4.14.23 **Evaluated By:** WDC

Client: City of Cadillac **PM:** BJH

Lab Notified of Rush or Short Holds: YES ~~NO~~

WO#: 50342119
PM: BJH Due Date: 04/28/23
CLIENT: GR-Cadillac

Project Received Via: FedEx ~~UPS~~ Client Pace Courier Other: ________

				Comments:
Custody Seal Present and Intact:	YES	NO	~~NA~~	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	~~NA~~	
Short Hold Present (≤ 48 Hours):	YES	~~NO~~		
Sample Received in Hold:	~~YES~~	NO		
Custody Signature Present:	~~YES~~	NO		
Collector Signature Present:	~~YES~~	NO		
Sample Collected Today and On Ice:	YES	NO	~~N/A~~	
IR Gun #: ~~350~~ 351	**Temp. should be 0°C - 6°C (Initial/Corrected)**			
Ice Type: WET Bagged / ~~WET Loose~~ BLUE NONE	**1.** Cooler Temp. Upon Receipt: 1.5/1.8 °C			
Ice Location: TOP BOTTOM MIDDLE ~~DISPERSED~~	**2.** Cooler Temp. Upon Receipt: ________ °C			
Temp Blank Received:	~~YES~~	NO		
Sample Label Matches COC (ID/Date/Time):	~~YES~~	NO		
Container Intact:	~~YES~~	NO		
Correct Container:	~~YES~~	NO		
Sufficient Volume:	~~YES~~	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation **pH Strip Lot #:** ________ *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	YES	NO	~~N/A~~	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** ________ *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable Cyanide*	YES	NO	~~N/A~~	
VOA Headspace Acceptable (<6mm):	YES	NO	~~N/A~~	
Trip Blank Received: HCl MeOH Other: ________	YES	~~NO~~	ON HOLD	

Comments:

3. Cooler Temp. Upon Receipt: ________ °C

4. Cooler Temp. Upon Receipt: ________ °C

Non-Conformance Form Required: YES ~~NO~~